

Inverse Design of Light-Trapping Layers

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Abstract

This report details the surrogate optimization and active learning approaches used to inversely design light-trapping layers.

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1 Introduction

Introduce the problem of light-trapping layers here [1].

2 Methods and theory

2.1 Rigorous Coupled Wave Analysis

This subsection will introduce the concept of Rigorous Coupled Wave Analysis (RCWA), a method in numerical photonics suited well for stacked periodic structures in the x and y axis with a period Λ_x, Λ_y respectively, and where each layer is homogeneous in the z -direction [3]. The structure's optical properties are completely described by the relative permittivity ϵ and relative permeability μ of each layer, meaning that each structure will have a geometry described by

$$\epsilon(x, y, z)_N = \epsilon(x + m \cdot \Lambda_x, y + n \cdot \Lambda_y)_N, m, n \in \mathbb{Z} \quad (1)$$

and

$$\mu(x, y, z)_N = \mu(x + m \cdot \Lambda_x, y + n \cdot \Lambda_y)_N, m, n \in \mathbb{Z}, \quad (2)$$

where m and n are integers and the subscript N denotes the N 'th layer of the structure. For the entirety of this project, the designed fourier patterned light trapping layers will be designed using crystalline silicon (Si), Titanium Dioxide (TiO_2) and Silicon Nitride (Si_3N_4), which are all nonmagnetic, so the relative permeability $\mu = 1$. In general, ϵ and μ will be 3×3 tensors, but for isotropic and linear materials this is a bit simpler. The equations solved for are of course the Maxwell equations, namely Faraday's law

$$\nabla \times \mathbf{E} = -\mu_0 \mu(x, y) \frac{\partial \mathbf{H}}{\partial t} \quad (3)$$

and Ampere's law

$$\nabla \times \mathbf{H} = \epsilon_0 \epsilon(x, y) \frac{\partial \mathbf{E}}{\partial t}, \quad (4)$$

Where ϵ_0 and μ_0 are the vacuum permittivity and vacuum permeability respectively. The structures are assumed free of charge and current,

$$\nabla \cdot (\epsilon \mathbf{E}) = 0$$

and

$$\nabla \cdot (\mu \mathbf{H}) = 0.$$

At a single frequency, the time dependency of Maxwell's equations may be solved by plane waves, see [2] section 9.2.1 for example, meaning

$$\mathbf{E} = \mathbf{E}_0 \exp i(\mathbf{k} \cdot \mathbf{r} - \omega t) \quad (5)$$

and

$$\mathbf{H} = \mathbf{H}_0 \exp i(\mathbf{k} \cdot \mathbf{r} - \omega t), \quad (6)$$

Where $\mathbf{E}_0$ and $\mathbf{H}_0$ are the (complex) amplitudes of the fields, with the physical fields being the real parts of $\mathbf{E}$ and $\mathbf{H}$, $\mathbf{k}$ being the wavevector, i.e. the spatial frequency, and ω being the frequency of the fields, also given by the dispersion relation $\omega = \frac{c|\mathbf{k}|}{n(\lambda)}$, where λ is the wavelength of the light.

Inserting equations 5 and 6 in equations 3 and 4 converts the time derivative to an algebraic factor of $-i\omega t$, meaning that the equations appear in their time harmonic form, common to many frequency domain numerical methods in photonics,

$$\nabla \times \mathbf{E} = i\omega \mu_0 \mu(x, y) \mathbf{H} \quad (7)$$

$$\nabla \times \mathbf{H} = -i\omega\mu_0\mu(x, y)\mathbf{E}. \quad (8)$$

Owing to the periodicity of the system, see equations 1 and 2, both the electromagnetic fields and the permittivity and permeability may be written as fourier series in the x- and y-directions as below,

$$\epsilon(x, y) = \sum_{m=-M}^M \sum_{n=-N}^N \epsilon_{m,n} \exp i(mL_x x + nL_y y), \quad (9)$$

$$\mu(x, y) = \sum_{m=-M}^M \sum_{n=-N}^N \mu_{m,n} \exp i(mL_x x + nL_y y), \quad (10)$$

$$\mathbf{E} = \exp i\mathbf{k} \cdot \mathbf{r} \sum_{m=-M}^M \sum_{n=-N}^N \mathbf{E}_{m,n} \exp i(mL_x x + nL_y y), \quad (11)$$

$$\mathbf{H} = \exp i\mathbf{k} \cdot \mathbf{r} \sum_{m=-M}^M \sum_{n=-N}^N \mathbf{H}_{m,n} \exp i(mL_x x + nL_y y), \quad (12)$$

where $L_x = \frac{2\pi}{\Lambda_x}$ and $L_y = \frac{2\pi}{\Lambda_y}$, and the parameters $\epsilon_{m,n}, \mu_{m,n}, \mathbf{E}_{m,n}$ and $\mathbf{H}_{m,n}$ are the fourier amplitudes of the m, n 'th diffraction order.

2.2 Machine Learning Surrogates

2.3 Optimization Strategies

3 Results

Present the performance of the surrogate models and the final inverse design outcomes.

Figure 1: Placeholder for a results figure.

4 Discussion

Discuss the results and what worked/didn't work.

5 Outlook

Discuss future work.

6 Conclusion

Summarize the findings.

References

- [1] A. Author and B. Writer. A study on light-trapping nanostructures. *Journal of Nanophotonics*, 10:1–10, 2026.
- [2] David J. Griffiths. *Introduction to Electrodynamics, fourth edition*. Cambridge University Press, 2017.
- [3] Changhyun Kim and Byoungcho Lee. Torcwa: Gpu-accelerated fourier modal method and gradient-based optimization for metasurface design. *Computer Physics Communications*, 282:108552, 2023.